AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2013

General Mathematics Paper II

Time: 2 hours 20 minutes Marks: 45

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Choices are specified inside the paper

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use colour pencils.
   DO NOT use staples, paper clips, glue or correcting fluid.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

If $\sqrt{y} \propto \sqrt{x}$ and $y = x = a^2$, then find the equation connecting x and y.

(ATTEMPT EITHER PART a OR PART b OF Q.2.)

Q.2. (Total 4 Marks)

a. The total amount of *Zakat* paid by two brothers Ahmar and Bilal is Rs. 50,000. Ahmar paid *Zakat* of Rs. 23,000 on his yearly savings. Find (4 Marks)

i. the amount of *Zakat* paid by Bilal
ii. the yearly savings of Ahmar and Bilal
iii. the ratio of *Zakat* paid by Ahmar and Bilal

b. Rehmani Sahib possesses a property worth Rs. 5, 00,000 and wishes to distribute his property among his legal heirs that include 1 son and 3 daughters. The son gets twice as much as a daughter gets. Find the amount for each legal heir. (4 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.3.)

Q.3. (Total 5 Marks)

a. Given that:

i. If profit $= x\%$, cost price $= y$, then find an expression for profit in terms of x and y. Hence, find an expression for the selling price. (3 Marks)

ii. If discount $= 5\%$, marked price $= z$, then find an expression for discount in terms of z. (2 Marks)

b. If the cost price, discount and marked price of an article are x, $\frac{8}{100}x$ and $\frac{118}{100}x$ respectively, then find the selling price of the article and show that the profit is 10 %. (5 Marks)

Q.4. (Total 3 Marks)

If an amount of Rs. 2, 00,000 is invested for 36 months at the rate of 8 percent per annum. Find the maturity value of the investment.

Q.5. (Total 5 Marks)

i. Define property tax. (1 Mark)

ii. Three partners, Mr. Aqeel, Mr. Kamran and Mr. Kashif, rented out 1 cottage and 2 shops. Given that the total income per month is 73,000, find the total income per year of each partner if each partner gets an equal share. Also find the tax rate if they collectively paid a property tax of Rs. 70,080. (4 Marks)

(ATTEMPT EITHER PART a OR PART b OF Q.6.)

Q.6. (Total 5 Marks)

a. If $\log_a \sqrt{(m - 2)} = 1$, then find m in terms of a. Hence find a if $m = 6$, where $a > 0$.

b. Evaluate $\sqrt[3]{64} + \frac{64}{\sqrt{64}} \times \sqrt[3]{10^6} - \frac{1}{(4)^{-1}}$

Q.7. (Total 4 Marks)

In a geometric progression if $T_n = \left(\frac{1}{2}\right)^{10}$ and $T_{n-1} = \left(\frac{1}{2}\right)^{9}$, then find the number of terms and the common ratio. Hence, find T_2.

(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 5 Marks)

a. Given that $A = \{x \mid x \text{ is a set of integers greater than or equal to zero}\}$.

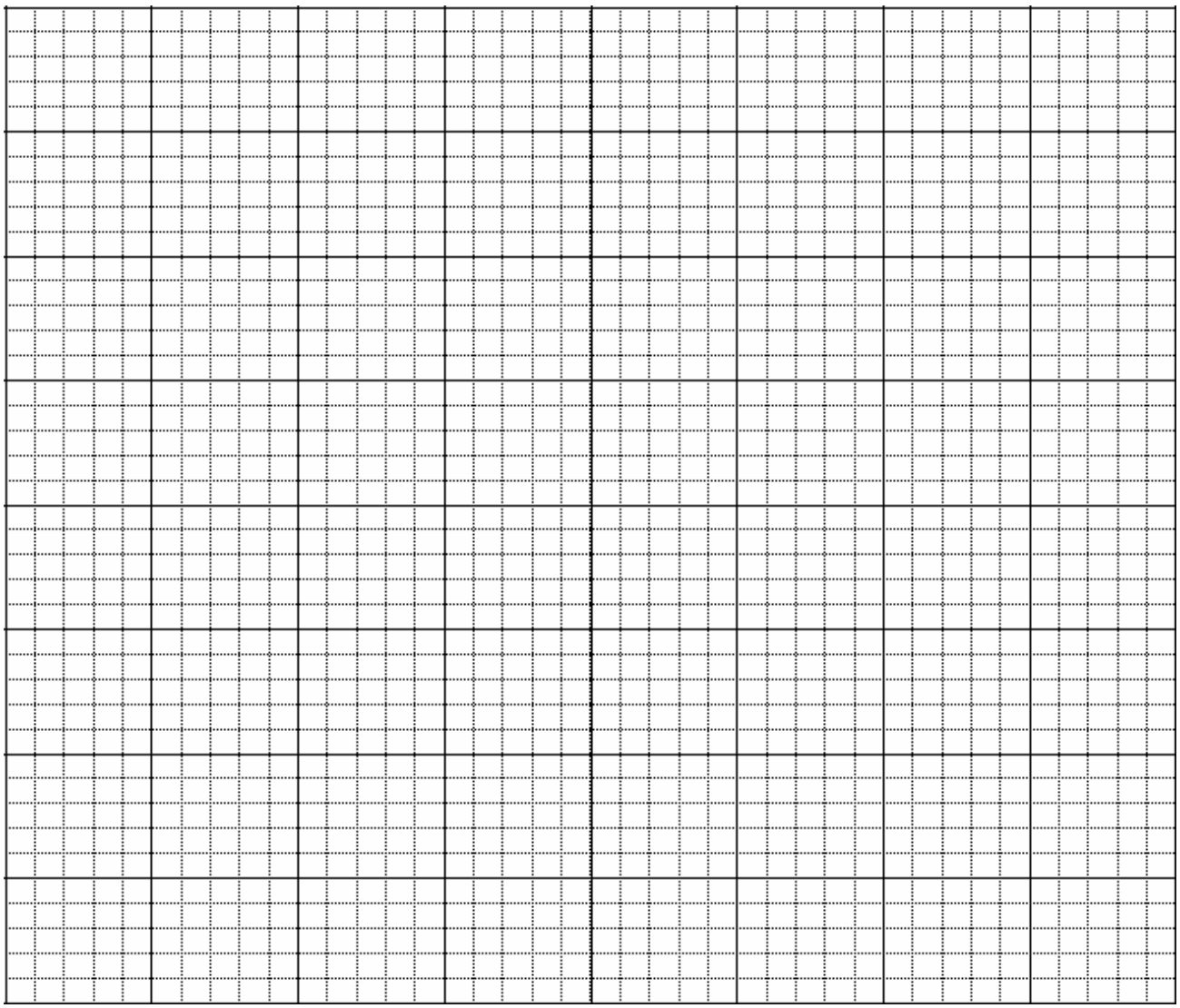
Write sets *B* and *C* whose elements are $\frac{x}{2}$ and x^2 respectively, where $x \in A$.

Also find $A \cup B$ and $A \cap B$.

b. If $U = \{0, \{\phi\}, \phi\}$, $T = \{0, \phi\}$ and $R = \{\phi\}$, then find $[T \cup (T - R)] \cap (R - T)$ and T^c.

Q.9. (Total 5 Marks)

Show graphically the given points $(1, -1)$, $(2, 1)$, $(4, -1)$ and $(5, 1)$ are the vertices of a parallelogram. Also find the coordinates of point of intersection of the diagonals.



(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 6 Marks)

a. Fill the missing entries in the given table. Hence, find the arithmetic mean for the given data.

Marks Obtained	**Number of Candidates**	**Class Mark (x)**	$f\,x$
1 – 5	9	3	
6 – 10	13	8	104
			195
		18	
21 – 25	27		621
Total	**100**		**1595**

b. Draw a histogram for the given data. Also locate mode on the histogram.

Class Interval	Frequency
2 – 7	4
8 – 13	5
14 – 19	7
20 – 25	8
26 – 31	6
32 – 37	3

END OF PAPER

Please use this page for rough work

Please use this page for rough work

Please use this page for rough work